

Schwartz Chapter 29: Weak interactions

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Quantum Field Theory

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1. Problem 29.1: Higgs production at LEP

The dominant production mechanism for Higgs bosons at LEP was $e^+e^- \rightarrow ZH$. Calculate the total cross section for this process at tree-level in the Standard Model. How many 100 GeV Higgs bosons would there have been when LEP ran at 206 GeV?

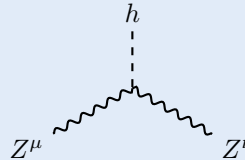
Solution. In this question, the relevant sector of the electroweak Lagrangian at tree-level is given by

$$\mathcal{L} = -\frac{1}{2}h(\Box + m_h^2)h + \bar{\psi}(i\not{\partial} - m_e)\psi + \frac{1}{v}m_Z^2 Z_\mu Z^\mu h + \frac{e}{\sin\theta_w \cos\theta_w} Z_\mu (J^{3\mu} - \sin^2\theta_w J^{\text{EM}\mu}), \quad (1)$$

where $v = 2\frac{m_W}{e} \sin\theta_w$, $J_\mu^{\text{EM}} = -\bar{\psi}\gamma_\mu\psi$ and

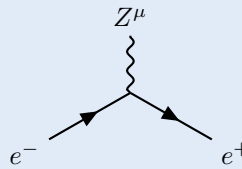
$$J_\mu^3 = \bar{\psi}^L \gamma_\mu T^3 \psi^L = \bar{\psi} P_R \gamma_\mu T^3 P_L \psi = \bar{\psi} \gamma_\mu P_L T^3 P_L \psi = \bar{\psi} \gamma_\mu T^3 P_L \psi = -\frac{1}{2} \bar{\psi} \gamma_\mu P_L \psi, \quad (2)$$

where we have used $[P_L, T^3] = 0$, $T^3 \psi^L = -\frac{1}{2} \psi^L$ and $P_L^2 = P_L$. Therefore, the ZZh interaction is given by



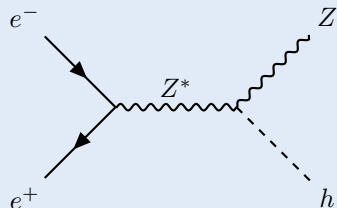
$$= i \frac{e}{\sin\theta_w \cos^2\theta_w} m_W g^{\mu\nu}. \quad (3)$$

The eeZ interaction is given by



$$= i \frac{e}{\sin\theta_w \cos\theta_w} \gamma^\mu \left(-\frac{1-\gamma^5}{4} + \sin^2\theta_w \right). \quad (4)$$

The only diagram we are concerned with at tree-level is



$$= i\mathcal{M}. \quad (5)$$

We can read off from the Feynman rules we just wrote down that

$$\begin{aligned}
 i\mathcal{M} &= \left(i \frac{e}{\sin \theta_w \cos \theta_w} \right) \left(i \frac{em_W}{\sin \theta_w \cos^2 \theta_w} \right) \\
 &\times \left[\bar{v}(p_2) \gamma^\mu \left(-\frac{1-\gamma^5}{4} + \sin^2 \theta_w \right) u(p_1) \right] \frac{-ig_{\mu\nu}}{s-m_Z^2} g^{\nu\rho} \epsilon_\rho^*(p_3) \\
 &= \frac{ie^2 m_W}{\sin^2 \theta_w \cos^3 \theta_w} \frac{1}{s-m_Z^2} \epsilon_\mu^*(p_3) \left[\bar{v}(p_2) \gamma^\mu \left(-\frac{1-\gamma^5}{4} + \sin^2 \theta_w \right) u(p_1) \right].
 \end{aligned} \tag{6}$$

Here the longitudinal part of the internal Z propagator vanishes by current conservation when $m_e = 0$. Taking the modulus square and summing over the external spins and polarisations, we find

$$\begin{aligned}
 \sum_{\text{spins/pols.}} |\mathcal{M}|^2 &= \left(\frac{e^2 m_W}{\sin^2 \theta_w \cos^3 \theta_w} \right)^2 \frac{1}{(s-m_Z^2)^2} \sum_{\text{pols.}} \epsilon_\mu^*(p_3) \epsilon_\nu(p_3) \\
 &\times \text{Tr} \left[\not{p}_2 \gamma^\mu \left(-\frac{1-\gamma^5}{4} + \sin^2 \theta_w \right) \not{p}_1 \gamma^\nu \left(-\frac{1-\gamma^5}{4} + \sin^2 \theta_w \right) \right],
 \end{aligned} \tag{7}$$

where we have taken $m_e = 0$. The symmetric part of the trace over spinors is

$$\left(4 \sin^4 \theta_w - 2 \sin^2 \theta_w + \frac{1}{2} \right) (p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1 \cdot p_2 g^{\mu\nu}). \tag{8}$$

The antisymmetric part does not contribute when contracted with the polarisation sum, which is given by

$$\sum_{\text{pols.}} \epsilon_\mu^*(p_3) \epsilon_\nu(p_3) = -g_{\mu\nu} + \frac{p_{3\mu} p_{3\nu}}{m_Z^2}. \tag{9}$$

Putting them together, we find

$$\begin{aligned}
 \sum_{\text{spins/pols.}} |\mathcal{M}|^2 &= \left(\frac{e^2 m_W}{\sin^2 \theta_w \cos^3 \theta_w} \right)^2 \frac{4 \sin^4 \theta_w - 2 \sin^2 \theta_w + \frac{1}{2}}{(s-m_Z^2)^2} \\
 &\times \left(2p_1 \cdot p_2 + \frac{2(p_1 \cdot p_3)(p_2 \cdot p_3)}{m_Z^2} - \frac{(p_1 \cdot p_2)p_3^2}{m_Z^2} \right).
 \end{aligned} \tag{10}$$

To compute the cross-section, we need to average over the initial spins, which is to multiply by a factor of $1/4$. In the centre-of-mass frame, we have

$$\begin{aligned}
 p_3^2 &= m_Z^2, \quad p_1 \cdot p_2 = \frac{s}{2}, \quad |\mathbf{k}| = \frac{\sqrt{s}}{2}, \\
 (p_1 \cdot p_3)(p_2 \cdot p_3) &= \frac{s}{4} (E_Z^2 - |\mathbf{p}|^2 \cos^2 \theta), \quad E_Z = \sqrt{|\mathbf{p}|^2 + m_Z^2},
 \end{aligned} \tag{11}$$

where $\mathbf{k}$ is the momentum of the incoming electron and positron and $\mathbf{p}$ is the momentum of the outgoing Z and Higgs bosons. Hence,

$$\begin{aligned}
 \sum_{\text{spins/pols.}} |\mathcal{M}|^2 &= \left(\frac{e^2 m_W}{\sin^2 \theta_w \cos^3 \theta_w} \right)^2 \frac{s (4 \sin^4 \theta_w - 2 \sin^2 \theta_w + \frac{1}{2})}{(s-m_Z^2)^2} \\
 &\times \left[1 + \frac{[s - (m_Z + m_h)^2][s - (m_Z - m_h)^2]}{8sm_Z^2} \sin^2 \theta \right].
 \end{aligned} \tag{12}$$

The magnitude of the outgoing three-momentum is

$$|\mathbf{p}| = \frac{\sqrt{[s - (m_Z + m_h)^2][s - (m_Z - m_h)^2]}}{2\sqrt{s}}, \quad (13)$$

and therefore

$$\frac{|\mathbf{p}|}{|\mathbf{k}|} = \frac{\sqrt{[s - (m_Z + m_h)^2][s - (m_Z - m_h)^2]}}{s}. \quad (14)$$

It follows that

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{1}{64\pi^2 s} \frac{|\mathbf{p}|}{|\mathbf{k}|} \frac{1}{4} \sum_{\text{spins/pols.}} |\mathcal{M}|^2 \\ &= \frac{\alpha^2 m_W^2 (4 \sin^4 \theta_w - 2 \sin^2 \theta_w + \frac{1}{2})}{16 \sin^4 \theta_w \cos^6 \theta_w (s - m_Z^2)^2} \frac{\sqrt{[s - (m_Z + m_h)^2][s - (m_Z - m_h)^2]}}{s} \\ &\quad \times \left[1 + \frac{[s - (m_Z + m_h)^2][s - (m_Z - m_h)^2]}{8sm_Z^2} \sin^2 \theta \right]. \end{aligned} \quad (15)$$

Using $\int d\Omega = 4\pi$ and $\int d\Omega \sin^2 \theta = 8\pi/3$, the total cross section is

$$\begin{aligned} \sigma(e^+e^- \rightarrow Zh) &= \frac{\pi \alpha^2 m_W^2 (4 \sin^4 \theta_w - 2 \sin^2 \theta_w + \frac{1}{2})}{4 \sin^4 \theta_w \cos^6 \theta_w (s - m_Z^2)^2} \frac{\sqrt{[s - (m_Z + m_h)^2][s - (m_Z - m_h)^2]}}{s} \\ &\quad \times \left[1 + \frac{[s - (m_Z + m_h)^2][s - (m_Z - m_h)^2]}{12sm_Z^2} \right]. \end{aligned} \quad (16)$$

For the numerical estimate, we take

$$\sqrt{s} = 206 \text{ GeV}, \quad m_h = 100 \text{ GeV}, \quad m_Z = 91.2 \text{ GeV}, \quad m_W = 80.4 \text{ GeV}, \quad (17)$$

together with

$$\sin^2 \theta_w = 0.231, \quad \alpha = \frac{1}{128}. \quad (18)$$

Substituting these values into the total cross section gives

$$\sigma(e^+e^- \rightarrow Zh) \simeq 1.09 \times 10^{-9} \text{ GeV}^{-2}. \quad (19)$$

Using

$$1 \text{ GeV}^{-2} = 3.894 \times 10^8 \text{ pb}, \quad (20)$$

this becomes

$$\sigma(e^+e^- \rightarrow Zh) \simeq 0.42 \text{ pb}. \quad (21)$$

The four LEP experiments collected an integrated luminosity of approximately 536 pb^{-1} at centre-of-mass energies of 206 GeV or higher. The expected number of produced Higgs bosons is therefore

$$N = \sigma L_{\text{int}} \simeq (0.42 \text{ pb})(536 \text{ pb}^{-1}) \simeq 2.3 \times 10^2. \quad (22)$$

Thus, approximately

$$\boxed{N \simeq 230} \quad (23)$$

100 GeV Higgs bosons would have been produced.

2. Problem 29.2: $e^+e^- \rightarrow \text{hadrons}$

$e^+e^- \rightarrow \text{hadrons}$ in the Standard Model.

1. Calculate the rate for the total cross section $\sigma_{\text{tot}}(e^+e^- \rightarrow \text{hadrons})$ in the Standard Model at tree-level including both Z -boson and photon contributions and their interference. The contribution using photons alone was calculated in Section 26.3.
2. Calculate σ_{tot} at 1-loop.
3. Plot the total cross section as a function of center-of-mass energy showing separately the photon contribution, the Z -boson contribution, and their sum. Plot also the sum ignoring interference between the Z -boson and photon contributions. When can interference be ignored?

Solution. For the $e^+e^- \rightarrow \gamma^* \rightarrow \text{hadrons}$, the relevant sector of the electroweak Lagrangian is

$$\mathcal{L} = \bar{\psi}_e(i\not{\partial} - m_e)\psi_e + \sum_i \bar{\psi}_i(i\not{\partial} - m_i)\psi_i - e\bar{\psi}_e\not{A}\psi_e + e \sum_i Q_i \bar{\psi}_i\not{A}\psi_i, \quad (1)$$

where i denotes the up, down, strange, etc. We can read off the scattering amplitude directly for a particular pair of quark and anti-quark, which is

$$i\mathcal{M}_\gamma^i = (ie)^2 Q_i [\bar{v}(p_2)\gamma^\mu u(p_1)] \frac{-i}{s} [\bar{u}_i(p_3)\gamma_\mu v_i(p_4)] = iQ_i \mathcal{M}_0, \quad (2)$$

where $\mathcal{M}_0$ is the scattering amplitude of $e^-e^+ \rightarrow \mu^-\mu^+$ at tree-level. Note that $Q_{u,c,t} = \frac{2}{3}$ and $Q_{d,s,b} = -\frac{1}{3}$.

The Z boson coupling to a general quark field is given by

$$\frac{ie}{\sin\theta_w \cos\theta_w} \gamma^\mu \left(T^3 \frac{1-\gamma^5}{2} - Q_i \sin^2\theta_w \right) = \frac{ie}{4\sin\theta_w \cos\theta_w} \gamma^\mu (\xi_i \gamma^5 - 4Q_i \sin^2\theta_w - \xi_i), \quad (3)$$

where

$$\xi_i = \begin{cases} -1, & i = u, c, t, \\ +1, & i = d, s, b. \end{cases} \quad (4)$$

Defining

$$a_i = 4|Q_i| \sin^2\theta_w - 1, \quad (5)$$

the quark vertex can equivalently be written as

$$\frac{ie}{4\sin\theta_w \cos\theta_w} \xi_i \gamma^\mu (\gamma^5 + a_i). \quad (6)$$

Therefore, the Z -exchange amplitude is

$$i\mathcal{M}_Z^i = \left(\frac{ie}{4\sin\theta_w \cos\theta_w} \right)^2 [\bar{v}(p_2)\gamma^\mu (\gamma^5 + a)u(p_1)] \frac{-i}{s - m_Z^2} [\xi_i \bar{u}(p_3)\gamma_\mu (\gamma^5 + a_i)v(p_4)], \quad (7)$$

where

$$a = 4\sin^2\theta_w - 1. \quad (8)$$

The initial-state spin sum is

$$\begin{aligned} & \sum_{\text{spins}} [\bar{v}(p_2)\gamma^\mu (\gamma^5 + a)u(p_1)] [\bar{u}(p_1)\gamma^\nu (\gamma^5 + a)v(p_2)] \\ &= \text{Tr} [\not{p}_2 \gamma^\mu (\gamma^5 + a) \not{p}_1 \gamma^\nu (\gamma^5 + a)] \\ &= 4(a^2 + 1) (p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1 \cdot p_2 \eta^{\mu\nu}) - 8ia p_{1\alpha} p_{2\beta} \epsilon^{\alpha\mu\beta\nu}. \end{aligned} \quad (9)$$

Similarly, since $\xi_i^2 = 1$, the final-state spin sum is

$$4(a_i^2 + 1) (p_{3\mu} p_{4\nu} + p_{3\nu} p_{4\mu} - p_3 \cdot p_4 \eta_{\mu\nu}) + 8ia_i p_{3\alpha} p_{4\beta} \epsilon^{\alpha}_{\mu}{}^{\beta}_{\nu}. \quad (10)$$

Thus,

$$\begin{aligned}
& \sum_{\text{spins}} |\mathcal{M}_Z^i|^2 \\
&= \left(\frac{e}{4 \sin \theta_w \cos \theta_w} \right)^4 \frac{1}{(s - m_Z^2)^2} \\
&\quad \times \left[4(a^2 + 1)(p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1 \cdot p_2 \eta^{\mu\nu}) - 8ia p_{1\alpha} p_{2\beta} \epsilon^{\alpha\mu\beta\nu} \right] \\
&\quad \times \left[4(a_i^2 + 1)(p_{3\mu} p_{4\nu} + p_{3\nu} p_{4\mu} - p_3 \cdot p_4 \eta_{\mu\nu}) + 8ia_i p_{3\alpha} p_{4\beta} \epsilon_{\mu}^{\alpha}{}^{\beta}{}_{\nu} \right].
\end{aligned} \tag{11}$$

The contraction between a symmetric tensor and an antisymmetric tensor vanishes. Contracting the two symmetric terms and the two epsilon tensors therefore gives

$$\begin{aligned}
& \sum_{\text{spins}} |\mathcal{M}_Z^i|^2 \\
&= \frac{e^4}{8 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \\
&\quad \times \left[((a^2 + 1)(a_i^2 + 1) - 4aa_i)(p_1 \cdot p_3)(p_2 \cdot p_4) \right. \\
&\quad \left. + ((a^2 + 1)(a_i^2 + 1) + 4aa_i)(p_1 \cdot p_4)(p_2 \cdot p_3) \right].
\end{aligned} \tag{12}$$

For massless external particles,

$$p_1 \cdot p_3 = p_2 \cdot p_4 = -\frac{t}{2}, \quad p_1 \cdot p_4 = p_2 \cdot p_3 = -\frac{u}{2}. \tag{13}$$

Hence,

$$\begin{aligned}
& \sum_{\text{spins}} |\mathcal{M}_Z^i|^2 \\
&= \frac{e^4}{32 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \\
&\quad \times \left[((a^2 + 1)(a_i^2 + 1) - 4aa_i)t^2 + ((a^2 + 1)(a_i^2 + 1) + 4aa_i)u^2 \right].
\end{aligned} \tag{14}$$

For the photon- Z interference term, the required mixed traces are

$$\begin{aligned}
& \text{Tr} \left[\not{p}_2 \gamma^\mu \not{p}_1 \gamma^\nu (\gamma^5 + a) \right] \\
&= 4a (p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1 \cdot p_2 \eta^{\mu\nu}) - 4ip_{1\alpha} p_{2\beta} \epsilon^{\alpha\mu\beta\nu}, \text{Tr} \left[\not{p}_3 \gamma_\mu \not{p}_4 \gamma_\nu (\xi_i \gamma^5 + \xi_i a_i) \right] \\
&= 4\xi_i a_i (p_{3\mu} p_{4\nu} + p_{3\nu} p_{4\mu} - p_3 \cdot p_4 \eta_{\mu\nu}) \\
&\quad + 4i\xi_i p_{3\alpha} p_{4\beta} \epsilon_{\mu}^{\alpha}{}^{\beta}{}_{\nu}.
\end{aligned} \tag{15}$$

Therefore,

$$\begin{aligned}
& 2 \text{Re} \sum_{\text{spins}} \mathcal{M}_\gamma^i \mathcal{M}_Z^{i\dagger} \\
&= \frac{e^4 Q_i \xi_i}{\sin^2 \theta_w \cos^2 \theta_w s (s - m_Z^2)} \\
&\quad \times \left[(aa_i - 1)t^2 + (aa_i + 1)u^2 \right].
\end{aligned} \tag{16}$$

Putting the photon, Z , and interference contributions together and averaging over the initial spins gives

$$\begin{aligned}
& \frac{1}{4} \sum_{\text{spin}} |\mathcal{M}^i|^2 \\
&= \frac{e^4 Q_i^2}{s^2} (t^2 + u^2) \\
&+ \frac{e^4}{128 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} [((a^2 + 1)(a_i^2 + 1) - 4aa_i)t^2 \\
&\quad + ((a^2 + 1)(a_i^2 + 1) + 4aa_i)u^2] \\
&+ \frac{e^4 Q_i \xi_i}{4 \sin^2 \theta_w \cos^2 \theta_w s(s - m_Z^2)} [(aa_i - 1)t^2 + (aa_i + 1)u^2].
\end{aligned} \tag{17}$$

Then, we sum over all flavours, which can be simplified as summing over the up and down flavour times three in our massless approximation. That is

$$\frac{1}{4} \sum_i \sum_{\text{spin}} |\mathcal{M}^i|^2 = \frac{3}{4} \sum_{\text{spin}} |\mathcal{M}^u|^2 + |\mathcal{M}^d|^2. \tag{18}$$

Using

$$Q_u = \frac{2}{3}, \quad Q_d = -\frac{1}{3}, \quad \xi_u = -1, \quad \xi_d = 1, \tag{19}$$

and

$$a_u = \frac{8}{3} \sin^2 \theta_w - 1, \quad a_d = \frac{4}{3} \sin^2 \theta_w - 1, \tag{20}$$

the required sums are

$$\begin{aligned}
& 3 \sum_{i=u,d} Q_i^2 \\
&= \frac{5}{3}, \\
& 3 \sum_{i=u,d} [(a^2 + 1)(a_i^2 + 1) - 4aa_i] \\
&= \frac{32}{3} \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23), \\
& 3 \sum_{i=u,d} [(a^2 + 1)(a_i^2 + 1) + 4aa_i] \\
&= \frac{16}{3} (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9), \\
& 3 \sum_{i=u,d} Q_i \xi_i (aa_i - 1) \\
&= \frac{8}{3} \sin^2 \theta_w (7 - 10 \sin^2 \theta_w), \\
& 3 \sum_{i=u,d} Q_i \xi_i (aa_i + 1) \\
&= -\frac{2}{3} (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9).
\end{aligned} \tag{21}$$

Therefore,

$$\begin{aligned}
& \frac{1}{4} \sum_i \sum_{\text{spins}} |\mathcal{M}^i|^2 \\
&= \frac{10e^4}{3s^2} (t^2 + u^2) \\
&+ \frac{e^4}{24 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) t^2 \right. \\
&\quad \left. + (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9) u^2 \right] \\
&+ \frac{e^4}{6 \sin^2 \theta_w \cos^2 \theta_w s (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) t^2 \right. \\
&\quad \left. - (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) u^2 \right].
\end{aligned} \tag{22}$$

One should also not forget to sum over all three outgoing colour states, namely $r\bar{r}$, $g\bar{g}$, and $b\bar{b}$, which gives another factor of 3. Therefore, the spin-averaged squared amplitude, summed over flavours and colours, is

$$\begin{aligned}
& \overline{|\mathcal{M}|^2} \\
&= \frac{10e^4}{s^2} (t^2 + u^2) \\
&+ \frac{e^4}{8 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) t^2 \right. \\
&\quad \left. + (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9) u^2 \right] \\
&+ \frac{e^4}{2 \sin^2 \theta_w \cos^2 \theta_w s (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) t^2 \right. \\
&\quad \left. - (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) u^2 \right].
\end{aligned} \tag{23}$$

In the centre-of-mass frame, we have

$$t = -\frac{s}{2}(1 - \cos \theta), \quad u = -\frac{s}{2}(1 + \cos \theta). \tag{24}$$

For massless two-body scattering,

$$\frac{d\sigma}{d\cos\theta} = \frac{1}{32\pi s} \overline{|\mathcal{M}|^2}. \tag{25}$$

Substituting t and u , we obtain

$$\begin{aligned}
& \frac{d\sigma}{d\cos\theta} \\
&= \frac{5e^4}{32\pi s} (1 + \cos^2 \theta) \\
&+ \frac{e^4 s}{1024\pi \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) (1 - \cos \theta)^2 \right. \\
&\quad \left. + (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9) (1 + \cos \theta)^2 \right] \\
&+ \frac{e^4}{256\pi \sin^2 \theta_w \cos^2 \theta_w (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) (1 - \cos \theta)^2 \right. \\
&\quad \left. - (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) (1 + \cos \theta)^2 \right].
\end{aligned} \tag{26}$$

Using

$$\int_{-1}^1 (1 \pm \cos \theta)^2 d\cos \theta = \frac{8}{3}, \tag{27}$$

the total cross section is

$$\begin{aligned}
 \sigma_{\text{tot}}(s) &= \frac{5e^4}{12\pi s} \\
 &+ \frac{e^4 s}{384\pi \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) \right. \\
 &\quad \left. + 80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9 \right] \\
 &+ \frac{e^4}{96\pi \sin^2 \theta_w \cos^2 \theta_w (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) \right. \\
 &\quad \left. - (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) \right].
 \end{aligned} \tag{28}$$

Taking

$$\alpha = \frac{e^2}{4\pi} = \frac{1}{128}, \quad \sin^2 \theta_w = 0.231, \quad m_Z = 91.2 \text{ GeV}, \tag{29}$$

and using $1 \text{ GeV}^{-2} = 3.894 \times 10^8 \text{ pb}$, this becomes

$$\sigma_{\text{tot}}(s) = \left[\frac{4.98 \times 10^5}{s} + \frac{2.93 \times 10^5 s}{(s - 8317)^2} - \frac{2.33 \times 10^4}{s - 8317} \right] \text{ pb}. \tag{30}$$

At one-loop order, we consider the leading QCD correction to the inclusive hadronic cross section. This should not be confused with the complete one-loop electroweak correction to $e^+e^- \rightarrow \text{hadrons}$.

The relevant contributions are the virtual-gluon correction to the outgoing quark current and the real-emission process

$$e^+e^- \rightarrow q\bar{q}g. \tag{31}$$

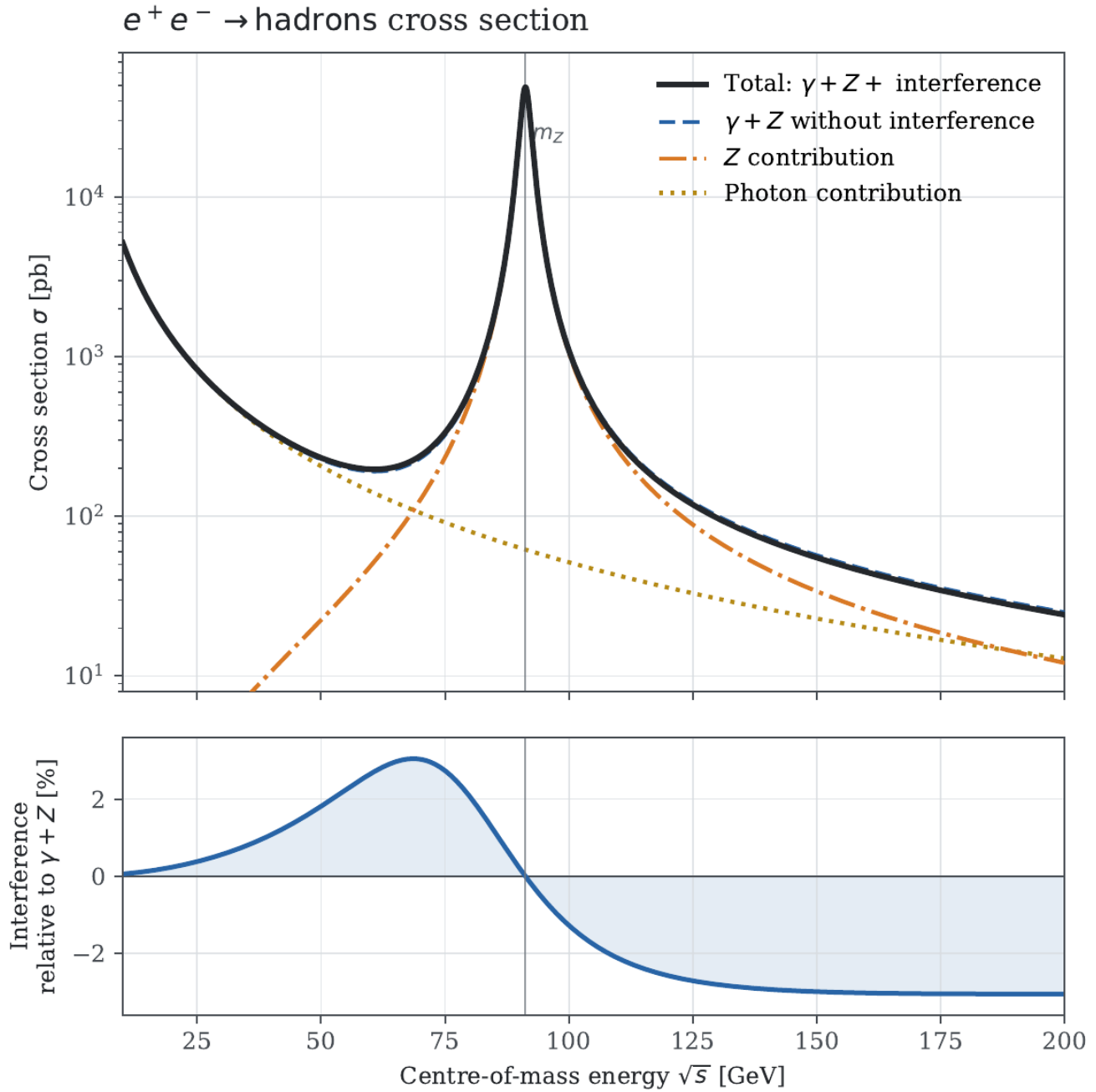
The virtual and real contributions are separately infrared divergent, but their divergences cancel in the inclusive cross section.

Because QCD couples vectorially to the quarks, the same correction applies to the vector and axial-vector parts of the quark current in the massless limit. It therefore multiplies the photon contribution, the Z -boson contribution, and their interference in the same way. Using the result of Section 26.3, we obtain

$$\begin{aligned}
 \sigma_{\text{NLO QCD}}(s) &= \sigma_{\text{tree}}(s) \left(1 + \frac{3C_F \alpha_s(s)}{4\pi} \right) + \mathcal{O}(\alpha_s^2) \\
 &= \left[\sigma_{\text{tree}}(s) \left(1 + \frac{\alpha_s(s)}{\pi} \right) \right] + \mathcal{O}(\alpha_s^2),
 \end{aligned} \tag{32}$$

where $C_F = 4/3$ and σ_{tree} contains the photon, Z -boson, and photon- Z interference contributions calculated in part (a).

This result includes the complete NLO QCD correction in the massless inclusive approximation. It does not include the full one-loop electroweak correction, which would additionally require electroweak self-energy, vertex, photon- Z mixing, box, counterterm, and real-photon contributions.



Six massless flavours, $N_c = 3$, $\alpha = 1/128$, $\sin^2 \theta_w = 0.231$, $m_Z = 91.2 \text{ GeV}$, $\Gamma_Z = 2.4952 \text{ GeV}$, $\alpha_s/\pi = 0.035$.

Figure 1. The NLO QCD cross section for $e^+ e^- \rightarrow \text{hadrons}$, assuming six massless quark flavours and $N_c = 3$. The lower panel shows the interference contribution relative to the incoherent sum $\sigma_\gamma + \sigma_Z$.

The interference can be neglected when $s \ll m_Z^2$, where photon exchange dominates, and close to the Z resonance, where Z exchange dominates. In particular, the interference vanishes at $s = m_Z^2$ because it is proportional to $s - m_Z^2$. Away from these regions, it produces a correction of a few percent and should generally be retained.

3. Problem 29.3: Higgs decays

Higgs decays.

1. Calculate the rate for $H \rightarrow b\bar{b}$ in the Standard Model.
2. Calculate the rate for $H \rightarrow gg$ in the Standard Model. The dominant contribution to this comes from a triangle loop diagram involving top quarks.
3. Calculate the rate for $H \rightarrow \gamma\gamma$ in the Standard Model. Include contributions both from top loops and from loops of W bosons.
4. Plot the branching ratios for $H \rightarrow b\bar{b}$, $H \rightarrow gg$, and $H \rightarrow \gamma\gamma$ as a function of Higgs mass.

Solution. Before symmetry breaking, the fermions couple to the Higgs field as

$$-Y_{ij}^d \bar{Q}^i H d_R^j - Y_{ij}^u \bar{Q}^i \tilde{H} u_R^j + h.c. \quad (1)$$

Focusing on the bottom quark, we find

$$-Y_{33}^d (\bar{t}_L \quad \bar{b}_L) \begin{pmatrix} H^+ \\ H^0 \end{pmatrix} b_R + h.c. \quad (2)$$

Expanding the Higgs around the vacuum as usual, we find the bottom quark couples to the Higgs field $h(x)$ through

$$-\frac{y}{\sqrt{2}} \bar{b}_L h b_R - \frac{y^*}{\sqrt{2}} \bar{b}_R h b_L = -\frac{1}{\sqrt{2}} (\text{Re } y) h \bar{b} b - \frac{i}{\sqrt{2}} (\text{Im } y) h \bar{b} \gamma^5 b, \quad (3)$$

where we defined $y \equiv Y_{33}^d$ and we have assumed y to be complex for generality. Therefore, at tree-level the scattering amplitude is given by

$$i\mathcal{M}_b = -\frac{1}{\sqrt{2}} i(\text{Re } y) [\bar{u}(p_2) v(p_3)] + \frac{1}{\sqrt{2}} (\text{Im } y) [\bar{u}(p_2) \gamma^5 v(p_3)]. \quad (4)$$

Then, we find

$$\begin{aligned} \sum_{\text{spin/colour}} |\mathcal{M}_b|^2 &= \frac{3}{2} \sum_{\text{spin}} \left[-(\text{Re } y) [\bar{u}(p_2) v(p_3)] - i(\text{Im } y) [\bar{u}(p_2) \gamma^5 v(p_3)] \right] \\ &\quad \times \left[-(\text{Re } y) [\bar{v}(p_3) u(p_2)] - i(\text{Im } y) [\bar{v}(p_3) \gamma^5 u(p_2)] \right] \\ &= \frac{3}{2} [(\text{Re } y)^2 + (\text{Im } y)^2] \text{Tr} \left((\not{p}_2 + m_b)(\not{p}_3 - m_b) \right) \\ &= 6|y|^2 (p_2 \cdot p_3 - m_b^2). \end{aligned} \quad (5)$$

In the mass basis, y can be chosen real. To compute the decay rate in the centre-of-mass frame, we use

$$\Gamma = \frac{1}{32\pi^2 m_h^2} |\mathbf{p}| \int d\Omega |\mathcal{M}_b|^2. \quad (6)$$

Using

$$s = m_h^2, \quad |\mathbf{p}| = \frac{1}{2} \sqrt{m_h^2 - 4m_b^2}, \quad p_2 \cdot p_3 = \frac{m_h^2 - 2m_b^2}{2}, \quad (7)$$

we find

$$\Gamma(H \rightarrow b\bar{b}) = \frac{3y^2}{16\pi m_h^2} (m_h^2 - 4m_b^2)^{3/2} = \frac{3m_b^2 m_h}{8\pi v^2} \left(1 - \frac{4m_b^2}{m_h^2} \right)^{3/2}. \quad (8)$$

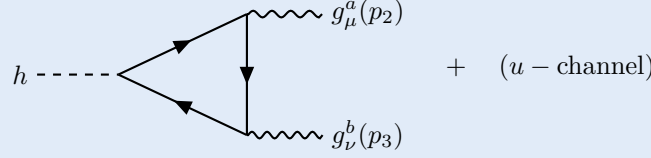
Using

$$m_h = 125.20 \text{ GeV}, \quad m_b = 4.183 \text{ GeV}, \quad v = 246.22 \text{ GeV}, \quad (9)$$

we obtain

$$\Gamma(H \rightarrow b\bar{b}) \simeq 4.28 \text{ MeV}. \quad (10)$$

There is no tree-level contribution to $H \rightarrow gg$. The lowest-order contribution comes from the quark triangle diagram



$$+ (u - \text{channel}). \quad (11)$$

For a particular quark flavour q , the scattering amplitude is

$$\begin{aligned} i\mathcal{M}_q^{ab} &= - \left(-i \frac{y_q}{\sqrt{2}} \right) (ig_s)^2 \int \frac{d^d k}{(2\pi)^d} \\ &\times \text{Tr} \left[\frac{i(\not{k} + m_q)}{k^2 - m_q^2} \gamma^\mu T^a \frac{i(\not{k} - \not{p}_2 + m_q)}{(k - p_2)^2 - m_q^2} \gamma^\nu T^b \frac{i(\not{k} - \not{p}_1 + m_q)}{(k - p_1)^2 - m_q^2} \right] \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \\ &+ (p_2, \mu, a \leftrightarrow p_3, \nu, b) \\ &= i \frac{y_q g_s^2}{\sqrt{2}} T_F \delta^{ab} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) 8m_q [F_q^{\mu\nu}(p_1, p_2) + F_q^{\nu\mu}(p_1, p_3)], \end{aligned} \quad (12)$$

where $y_q = \frac{\sqrt{2}m_q}{v} = 2\sqrt{\lambda} \frac{m_q}{m_h}$, and $F_q^{\mu\nu}$ is defined to include the factor of i below, so that it is real for the top-quark loop. Since $y_q \propto m_q$ and $m_b, m_c, m_s, m_d, m_u \ll m_h$, the dominant contribution comes from the top-quark loop. The complete amplitude is obtained by summing over quark flavours. Now, we evaluate the loop integral. The trace can be simplified as

$$\begin{aligned} &\text{Tr} \left[\frac{i(\not{k} + m_q)}{k^2 - m_q^2} \gamma^\mu \frac{i(\not{k} - \not{p}_2 + m_q)}{(k - p_2)^2 - m_q^2} \gamma^\nu \frac{i(\not{k} - \not{p}_1 + m_q)}{(k - p_1)^2 - m_q^2} \right] \\ &= - \frac{4im_q}{(k^2 - m_q^2)[(k - p_2)^2 - m_q^2][(k - p_1)^2 - m_q^2]} \left[k^\mu (k - p_2)^\nu + k^\nu (k - p_2)^\mu - k \cdot (k - p_2) \eta^{\mu\nu} \right. \\ &\quad + k^\mu (k - p_1)^\nu + k^\nu (k - p_1)^\mu + k \cdot (k - p_1) \eta^{\mu\nu} + (k - p_2)^\mu (k - p_1)^\nu + (k - p_1)^\mu (k - p_2)^\nu \\ &\quad \left. - (k - p_2) \cdot (k - p_1) \eta^{\mu\nu} + m_q^2 \eta^{\mu\nu} \right]. \end{aligned} \quad (13)$$

Let

$$A = k^2 - m_q^2, \quad B = (k - p_2)^2 - m_q^2, \quad C = (k - p_1)^2 - m_q^2. \quad (14)$$

Introducing Feynman parameter,

$$\frac{1}{ABC} = 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xA + yB + zC]^3}, \quad z = 1 - x - y, \quad (15)$$

the denominator becomes

$$\begin{aligned} xA + yB + zC &= k^2 - 2k \cdot (yp_2 + zp_1) + yp_2^2 + zp_1^2 - m_q^2 \\ &= (k - yp_2 - zp_1)^2 - \Delta, \end{aligned} \quad (16)$$

where

$$\Delta = m_q^2 - y(1 - y)p_2^2 - z(1 - z)p_1^2 + 2yz p_1 \cdot p_2. \quad (17)$$

Shifting the loop momentum, $\ell = k - yp_2 - zp_1 \equiv k - P$, the numerator becomes

$$\begin{aligned} \mathcal{N}^{\mu\nu} &= 4\ell^\mu \ell^\nu - \ell^2 \eta^{\mu\nu} + 4P^\mu P^\nu - 2P^\mu (p_1 + p_2)^\nu - 2P^\nu p_2^\mu \\ &\quad + p_1^\mu p_2^\nu + p_2^\mu p_1^\nu + (m_q^2 - P^2 + 2P \cdot p_2 - p_1 \cdot p_2) \eta^{\mu\nu} \\ &\quad + 4\ell^\mu P^\nu + 4P^\mu \ell^\nu - 2\ell^\mu (p_1 + p_2)^\nu - 2\ell^\nu p_2^\mu + 2\ell \cdot (p_2 - P) \eta^{\mu\nu}. \end{aligned} \quad (18)$$

The loop integral becomes

$$F_q^{\mu\nu}(p_1, p_2) = i \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d \ell}{(2\pi)^d} \frac{\mathcal{N}^{\mu\nu}(\ell, P)}{(\ell^2 - \Delta)^3}. \quad (19)$$

The terms with an odd number of ℓ in the numerator vanish, and $\ell^\mu \ell^\nu \rightarrow \frac{1}{d} \eta^{\mu\nu} \ell^2$, which gives

$$\begin{aligned} \mathcal{N}^{\mu\nu} \longrightarrow & \left(\frac{4}{d} - 1 \right) \ell^2 \eta^{\mu\nu} + 4P^\mu P^\nu - 2P^\mu (p_1 + p_2)^\nu - 2P^\nu p_2^\mu \\ & + p_1^\mu p_2^\nu + p_2^\mu p_1^\nu + (m_q^2 - P^2 + 2P \cdot p_2 - p_1 \cdot p_2) \eta^{\mu\nu}. \end{aligned} \quad (20)$$

Then, using dimensional regularisation expanding around $d = 4 - \epsilon$, we find

$$\begin{aligned} F_q^{\mu\nu}(p_1, p_2) = \frac{-1}{32\pi^2} \int_0^1 dx \int_0^{1-x} dy \frac{1}{m_q^2 - xz m_h^2} \Bigg[& \\ & m_h^2 \left(\frac{1}{2} - 2xz \right) \eta^{\mu\nu} \\ & - 2z(2z - 1) p_1^\mu p_1^\nu + 4y(1 - y) p_2^\mu p_2^\nu \\ & - (4yz - 2z + 1) p_1^\mu p_2^\nu \\ & - (4yz - 2y - 2z + 1) p_2^\mu p_1^\nu \Bigg]. \end{aligned} \quad (21)$$

Contracting with the on-shell gluon polarisations, it follows that

$$\begin{aligned} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) p_1^\mu p_1^\nu &= \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) p_1^\mu p_2^\nu \\ &= [\epsilon^*(p_2) \cdot p_3] [\epsilon^*(p_3) \cdot p_2], \end{aligned} \quad (22)$$

while all terms proportional to p_2^μ vanish. Therefore,

$$\begin{aligned} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) F_t^{\mu\nu}(p_1, p_2) &= -\frac{1}{32\pi^2} \int_0^1 dx \int_0^{1-x} dy \frac{1 - 4xz}{m_t^2 - xz m_h^2} \\ &\times \left[\frac{m_h^2}{2} \epsilon^*(p_2) \cdot \epsilon^*(p_3) - (\epsilon^*(p_2) \cdot p_3) (\epsilon^*(p_3) \cdot p_2) \right]. \end{aligned} \quad (23)$$

The crossed orientation gives the same result after relabelling the Feynman parameters. Adding the two orientations gives

$$\begin{aligned} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) [F_t^{\mu\nu}(p_1, p_2) + F_t^{\nu\mu}(p_1, p_3)] &= -\frac{1}{16\pi^2} \left[\frac{m_h^2}{2} \epsilon^*(p_2) \cdot \epsilon^*(p_3) - (\epsilon^*(p_2) \cdot p_3) (\epsilon^*(p_3) \cdot p_2) \right] \\ &\times \int_0^1 dx \int_0^{1-x} dy \frac{1 - 4x(1 - x - y)}{m_t^2 - m_h^2 x(1 - x - y)}. \end{aligned} \quad (24)$$

Evaluating the integral gives

$$\begin{aligned} \int_0^1 dx \int_0^{1-x} dy \frac{1 - 4x(1 - x - y)}{m_t^2 - m_h^2 x(1 - x - y)} &= \frac{2}{m_h^2} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right]. \end{aligned} \quad (25)$$

Thus, the top-loop amplitude becomes

$$\begin{aligned} i\mathcal{M}_t^{ab} &= -i \frac{y_t g_s^2 m_t T_F}{2\sqrt{2}\pi^2} \delta^{ab} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right] \\ &\times \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \left[\eta^{\mu\nu} - \frac{2}{m_h^2} p_3^\mu p_2^\nu \right]. \end{aligned} \quad (26)$$

Using

$$\sum_{\text{pols.}} \epsilon_\mu^*(p) \epsilon_\nu(p) = -\eta_{\mu\nu}, \quad \sum_{a,b} \delta^{ab} \delta^{ab} = N_c^2 - 1 = 8, \quad T_F = \frac{1}{2}, \quad (27)$$

and

$$\left[\eta^{\mu\nu} - \frac{2}{m_h^2} p_3^\mu p_2^\nu \right] \left[\eta_{\mu\nu} - \frac{2}{m_h^2} p_{3\mu} p_{2\nu} \right] = 2, \quad (28)$$

we finally obtain

$$\sum_{\text{pols./colours}} |\mathcal{M}_t|^2 = \frac{m_t^2 y_t^2 g_s^4}{2\pi^4} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right]^2. \quad (29)$$

Here we have neglected the contributions from the remaining quarks.

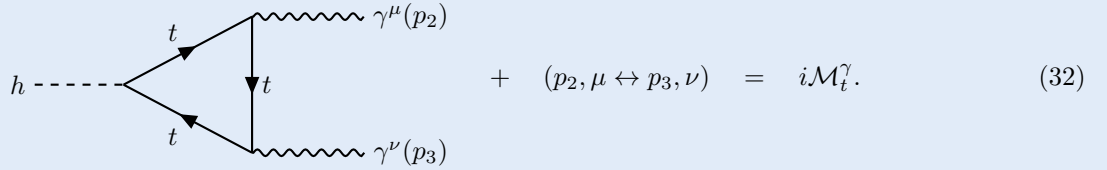
Plugging into (6), we find

$$\Gamma(H \rightarrow gg) = \frac{\lambda g_s^4 m_t^4}{16\pi^5 m_h^3} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right]^2, \quad (30)$$

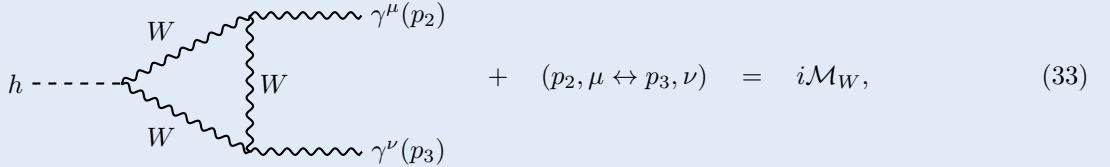
where a factor of 1/2 is added since the outgoing gluon states are identical. This gives a numerical value of

$$\Gamma(H \rightarrow gg) \simeq 1.97 \times 10^{-4} \text{ GeV} = 0.197 \text{ MeV}. \quad (31)$$

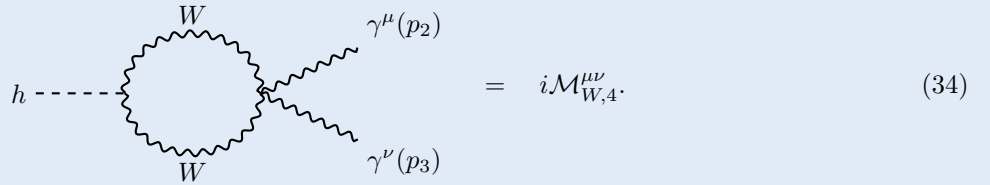
Now we turn to the computation of $H \rightarrow \gamma\gamma$. Similarly to the gluon case, there is no tree-level contribution to the process. The dominant quark-loop contribution comes from the top-quark loop, just as for the gluon case. On top of that, there is also a contribution from a W -boson loop. Therefore, our job is to compute the following diagrams



$$+ (p_2, \mu \leftrightarrow p_3, \nu) = i\mathcal{M}_t^\gamma. \quad (32)$$



$$+ (p_2, \mu \leftrightarrow p_3, \nu) = i\mathcal{M}_W, \quad (33)$$

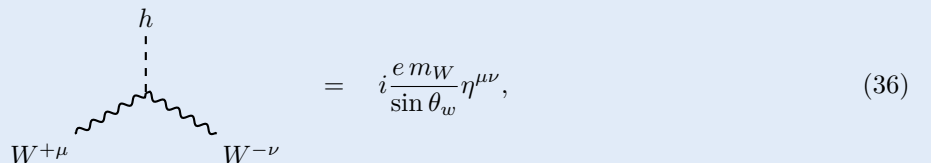


$$= i\mathcal{M}_{W,4}^{\mu\nu}. \quad (34)$$

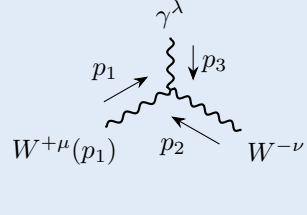
The relevant interactions are

$$\begin{aligned} \mathcal{L}_{\text{int}} = & -ie[F^{\mu\nu} W_\mu^+ W_\nu^- - (\partial_\mu W_\nu^+ - \partial_\nu W_\mu^+) A^\mu (W^-)^\nu + (\partial_\mu W_\nu^- - \partial_\nu W_\mu^-) A^\mu (W^+)^\nu] \\ & + 2\frac{h}{v} m_W^2 W_\mu^+ W_\mu^- + e^2 (A^\mu W_\mu^+ A^\nu W_\nu^- - A_\mu^2 W_\nu^+ (W^-)^\nu) - \frac{y_t}{\sqrt{2}} h \bar{t} t + \frac{2e}{3} A_\mu \bar{t} \gamma^\mu t, \end{aligned} \quad (35)$$

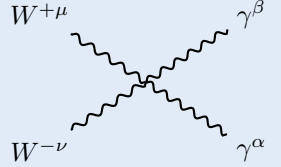
where the first three interactions are given by the diagrams



$$= i \frac{e m_W}{\sin \theta_w} \eta^{\mu\nu}, \quad (36)$$



$$= -ie \left[\eta^{\mu\nu}(p_1 - p_2)^\lambda + \eta^{\nu\lambda}(p_2 - p_3)^\mu + \eta^{\lambda\mu}(p_3 - p_1)^\nu \right], \quad (37)$$



$$= ie^2 \left[\eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu} - 2\eta^{\alpha\beta}\eta^{\mu\nu} \right]. \quad (38)$$

and the last two are Yukawa and QED interactions. Note that $i\mathcal{M}_t^\gamma$ is almost identical to $i\mathcal{M}_t$, but now the colour trace gives $N_c = 3$ and $g_s \rightarrow 2e/3$. Therefore, we can read off from (26),

$$i\mathcal{M}_t^\gamma = -i \frac{2y_t e^2 m_t}{3\sqrt{2}\pi^2} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right] \times \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \left[\eta^{\mu\nu} - \frac{2}{m_h^2} p_3^\mu p_2^\nu \right]. \quad (39)$$

Although rather tedious, we can also read off the W bosons loop diagram from the Feynman rules (or one can easily obtain the expression using the power of AI). In unitary gauge, the W -boson propagator is

$$D_W^{\alpha\beta}(q) = \frac{-iP^{\alpha\beta}(q)}{q^2 - m_W^2}, \quad P^{\alpha\beta}(q) \equiv \eta^{\alpha\beta} - \frac{q^\alpha q^\beta}{m_W^2}. \quad (40)$$

It is useful to define the momentum-dependent part of the $\gamma W^+ W^-$ vertex by

$$\Gamma^{\lambda\alpha\beta}(q, r, s) = \eta^{\alpha\beta}(r - s)^\lambda + \eta^{\beta\lambda}(s - q)^\alpha + \eta^{\lambda\alpha}(q - r)^\beta, \quad (41)$$

so that the vertex is $-ie\Gamma^{\lambda\alpha\beta}(q, r, s)$.

The triangle diagram and its crossed photon ordering together give

$$i\mathcal{M}_W = \frac{e^3 m_W}{\sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int \frac{d^d k}{(2\pi)^d} \left[\frac{\mathcal{N}^{\mu\nu}(k; p_2, p_3)}{(k^2 - m_W^2)[(k - p_2)^2 - m_W^2][(k - p_1)^2 - m_W^2]} + \frac{\mathcal{N}^{\nu\mu}(k; p_3, p_2)}{(k^2 - m_W^2)[(k - p_3)^2 - m_W^2][(k - p_1)^2 - m_W^2]} \right], \quad (42)$$

where

$$\mathcal{N}^{\mu\nu}(k; p_2, p_3) = \eta^{\alpha\rho} P_{\alpha\beta}(k) \Gamma^{\mu\beta\gamma}(-p_2, k, p_2 - k) \times P_{\gamma\delta}(k - p_2) \Gamma^{\nu\delta\sigma}(-p_3, k - p_2, p_1 - k) P_{\sigma\rho}(k - p_1). \quad (43)$$

The quartic $\gamma\gamma W^+ W^-$ bubble diagram must also be included. Define

$$Q^{\mu\nu\beta\sigma} = \eta^{\mu\beta}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\beta} - 2\eta^{\mu\nu}\eta^{\beta\sigma}, \quad (44)$$

so that the quartic vertex is $ie^2 Q^{\mu\nu\beta\sigma}$. Its contribution is

$$i\mathcal{M}_{W,4} = \frac{e^3 m_W}{\sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int \frac{d^d k}{(2\pi)^d} \frac{\eta^{\alpha\rho} P_{\alpha\beta}(k) Q^{\mu\nu\beta\sigma} P_{\sigma\rho}(k - p_1)}{(k^2 - m_W^2)[(k - p_1)^2 - m_W^2]}. \quad (45)$$

Now, we shall evaluate the two diagrams. Introduce Feynman parameters,

$$\begin{aligned} & \frac{1}{D_0 D_2 D_1} \\ &= 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xD_0 + yD_2 + zD_1]^3}, \quad \frac{1}{D_0 D_3 D_1} \\ &= 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xD_0 + yD_3 + zD_1]^3}, \quad z = 1 - x - y, \end{aligned} \quad (46)$$

where

$$\begin{aligned} D_0 &= k^2 - m_W^2, \\ D_1 &= (k - p_1)^2 - m_W^2, D_2 = (k - p_2)^2 - m_W^2, \\ D_3 &= (k - p_3)^2 - m_W^2. \end{aligned} \quad (47)$$

For the first and crossed orientations, respectively, make the shifts

$$\begin{aligned} \ell &= k - yp_2 - zp_1, \\ \ell &= k - yp_3 - zp_1. \end{aligned} \quad (48)$$

Using

$$p_1 = p_2 + p_3, \quad p_1^2 = m_h^2, \quad p_2^2 = p_3^2 = 0, \quad (49)$$

both denominators reduce to

$$xD_0 + yD_{2,3} + zD_1 = \ell^2 - \Delta, \quad \Delta = m_W^2 - xzm_h^2. \quad (50)$$

Therefore, the two triangle orientations together become

$$\begin{aligned} i\mathcal{M}_W &= 2 \frac{e^3 m_W}{\sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d \ell}{(2\pi)^d} \\ &\times \frac{\mathcal{N}^{\mu\nu}(\ell + yp_2 + zp_1; p_2, p_3) + \mathcal{N}^{\nu\mu}(\ell + yp_3 + zp_1; p_3, p_2)}{(\ell^2 - \Delta)^3}. \end{aligned} \quad (51)$$

For convenience, define the combined numerator

$$\begin{aligned} \tilde{\mathcal{N}}^{\mu\nu}(\ell; x, y) &\equiv \mathcal{N}^{\mu\nu}(\ell + yp_2 + zp_1; p_2, p_3) \\ &+ \mathcal{N}^{\nu\mu}(\ell + yp_3 + zp_1; p_3, p_2). \end{aligned} \quad (52)$$

Thus,

$$i\mathcal{M}_W = 2 \frac{e^3 m_W}{\sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d \ell}{(2\pi)^d} \frac{\tilde{\mathcal{N}}^{\mu\nu}(\ell; x, y)}{(\ell^2 - \Delta)^3}. \quad (53)$$

To ensure convergence we should include the four-point diagram before carrying out the loop integral. The terms with odd l in the numerator vanishes and $l^\mu l^\nu \rightarrow \frac{1}{d} \eta^{\mu\nu} l^2$, the l^4 term in $i\mathcal{M}_{W,4}$ becomes

$$\begin{aligned} \ell^\mu \ell^\nu \ell^\rho \ell^\sigma &\longrightarrow \frac{\ell^4}{d(d+2)} \left(\right. \\ &\left. \eta^{\mu\nu} \eta^{\rho\sigma} + \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho} \right). \end{aligned} \quad (54)$$

After some long and cumbersome algebra, one would find

$$\begin{aligned} i\mathcal{M}_W + i\mathcal{M}_{W,4} &= i \frac{e^3}{16\pi^2 m_W \sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) [(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu] \\ &\times \left[2 + 12m_W^2 \int_0^1 dx \int_0^{1-x} dy \frac{1 - 2xz}{m_W^2 - m_h^2 xz} \right]. \end{aligned} \quad (55)$$

The integral can be evaluated, which gives

$$\int_0^1 dx \int_0^{1-x} dy \frac{1 - 2xz}{m_W^2 - m_h^2 xz} = \frac{1}{m_h^2} \left[1 + \left(2 - \frac{4m_W^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_W} \right) \right]. \quad (56)$$

Finally, we end up with

$$\begin{aligned} i\mathcal{M}_W + i\mathcal{M}_{W,4} &= i \frac{e^3}{16\pi^2 m_W \sin \theta_w} \left[2 + \frac{12m_W^2}{m_h^2} + \frac{24m_W^2}{m_h^2} \left(1 - \frac{2m_W^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_W} \right) \right] \\ &\times \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) [(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu]. \end{aligned} \quad (57)$$

To combine the top and W contributions, define

$$\mathcal{F}_t = \frac{32m_t^2}{3m_h^2} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right], \mathcal{F}_W = 2 + \frac{12m_W^2}{m_h^2} + \frac{24m_W^2}{m_h^2} \left(1 - \frac{2m_W^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_W} \right). \quad (58)$$

Using $y_t = \sqrt{2}m_t/v$ and $m_W = ev/(2\sin\theta_w)$, the total amplitude is

$$i\mathcal{M}_{\gamma\gamma} = i \frac{e^3}{16\pi^2 m_W \sin\theta_w} (\mathcal{F}_W - \mathcal{F}_t) \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) [(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu]. \quad (59)$$

Thus, using $p_2 \cdot p_3 = m_h^2/2$, the polarization sum gives

$$\begin{aligned} \sum_{\text{pols.}} |\mathcal{M}_{\gamma\gamma}|^2 &= \frac{e^6}{256\pi^4 m_W^2 \sin^2\theta_w} (\mathcal{F}_W - \mathcal{F}_t)^2 [(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu] [(p_2 \cdot p_3) \eta_{\mu\nu} - p_{3\mu} p_{2\nu}] \\ &= \frac{e^6 m_h^4}{512\pi^4 m_W^2 \sin^2\theta_w} (\mathcal{F}_W - \mathcal{F}_t)^2. \end{aligned} \quad (60)$$

The two photons are identical, so the two-body phase space includes a factor of $1/2!$. Therefore,

$$\Gamma(H \rightarrow \gamma\gamma) = \frac{e^6 m_h^3}{16384\pi^5 m_W^2 \sin^2\theta_w} (\mathcal{F}_W - \mathcal{F}_t)^2 = \frac{\alpha^2 m_h^3}{256\pi^3 v^2} (\mathcal{F}_W - \mathcal{F}_t)^2. \quad (61)$$

Using

$$\begin{aligned} m_h &= 125.20 \text{ GeV}, \\ m_t &= 172.56 \text{ GeV}, \\ m_W &= 80.377 \text{ GeV}, v = 246.22 \text{ GeV}, \\ \alpha^{-1} &= 137.036, \end{aligned} \quad (62)$$

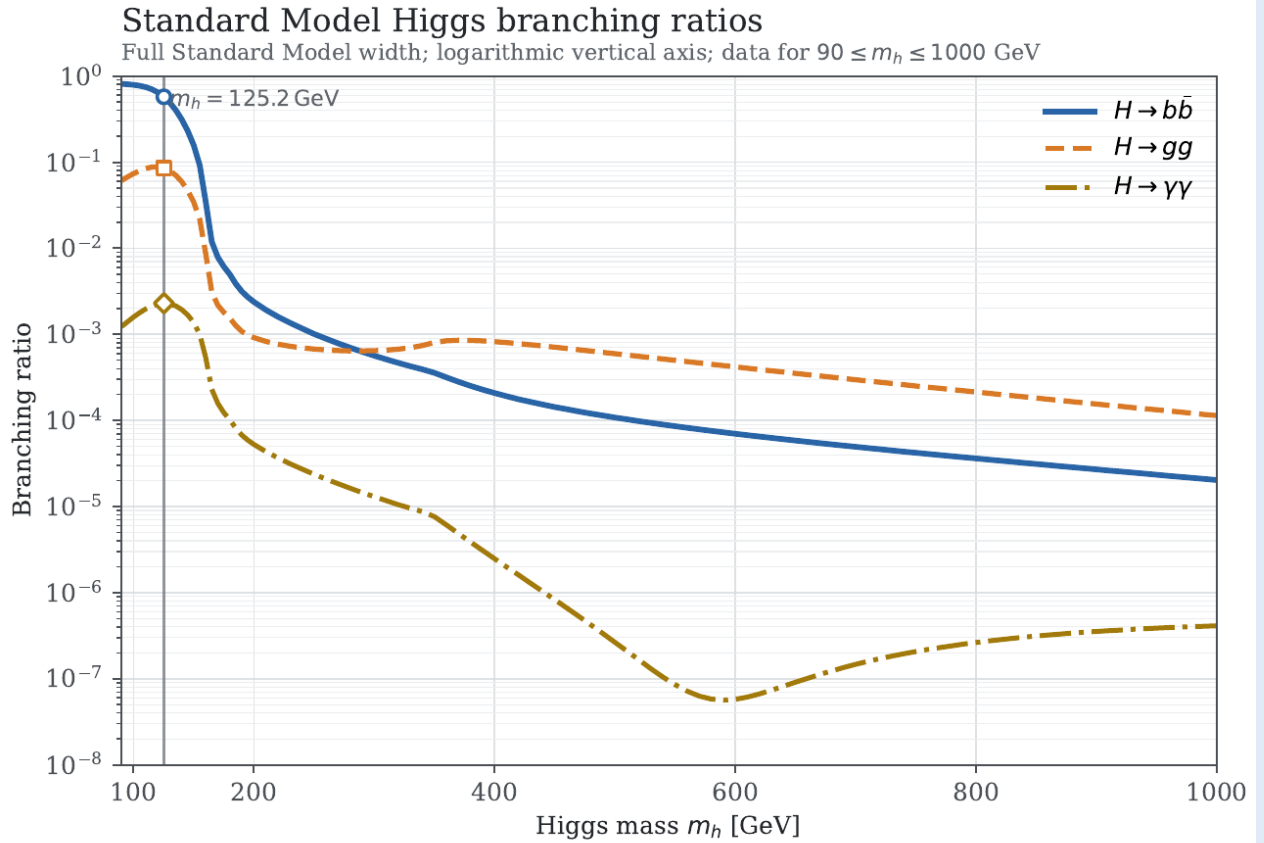
we find

$$\mathcal{F}_t \simeq 1.836, \quad \mathcal{F}_W \simeq 8.331, \quad (63)$$

and hence

$$\Gamma(H \rightarrow \gamma\gamma) \simeq 9.16 \times 10^{-6} \text{ GeV} = 9.16 \text{ keV}. \quad (64)$$

Finally, the three branching ratios are compared in Fig. 2. The denominator is the full Standard Model Higgs width, rather than only the sum of the three channels calculated above. The broad mass scan is taken from the LHC Higgs Cross Section Working Group report CERN-2011-002, while the precise values quoted below at $m_h = 125.2$ GeV use the newer Yellow Report 4 table.



Source: LHC Higgs Cross Section Working Group, CERN-2011-002, Tables 26-29.

Figure 2. Standard Model branching ratios for $H \rightarrow b\bar{b}$, $H \rightarrow gg$, and $H \rightarrow \gamma\gamma$ for $90 \leq m_h \leq 1000$ GeV. The values are taken from Tables 26–29 of CERN-2011-002 and use the full Standard Model decay width. Above approximately 500 GeV, the on-shell Higgs treatment used in these tables should be interpreted with care.

At $m_h = 125.2$ GeV, the corresponding values are

$$\text{BR}(H \rightarrow b\bar{b}) = 0.5792, \quad \text{BR}(H \rightarrow gg) = 0.08172, \quad \text{BR}(H \rightarrow \gamma\gamma) = 0.002270. \quad (65)$$

4. Problem 29.6: Phases in the PMNS matrix

Show that with general Dirac and Majorana mass matrices, there are three phases in the PMNS matrix, while if the mass matrix is purely Dirac, there is only one. How many phases are there if the masses are purely Majorana?

Solution. A general 3×3 unitary PMNS matrix contains three mixing angles and six phases. The three charged-lepton fields can always be rephased, removing three phases. If the neutrinos are Dirac particles, the three neutrino mass eigenstates can also be rephased. However, one common rephasing of all charged leptons and neutrinos leaves the PMNS matrix unchanged, so only $3 + 3 - 1 = 5$ phases can be removed, leaving one physical phase. If any Majorana mass is present, the neutrino mass eigenstates are generically Majorana and cannot be continuously rephased, since the Majorana condition $N_i^c = N_i$ is preserved only by the discrete transformation $N_i \rightarrow \pm N_i$. Consequently, only the three charged-lepton

rephasings are available, leaving $6 - 3 = 3$ physical phases. The same reasoning applies to purely Majorana masses. Thus the PMNS matrix contains one physical phase for purely Dirac neutrinos and three physical phases for either Dirac-plus-Majorana or purely Majorana neutrinos.

5. Problem 29.7: Neutrino oscillations

Neutrino oscillations.

1. Neutrinos are produced in the Sun predominantly through the reaction $p + p + e^- \rightarrow d + \nu_e$. What is the momentum of the neutrinos produced this way?
2. Consider a two-neutrino flavour system. The mass eigenstates evolve in time as

$$|\nu_1\rangle \quad (1)$$

$$= e^{-iE_1 t} (\cos \theta |\nu_e\rangle + \sin \theta |\nu_\mu\rangle), \quad (2)$$

$$|\nu_2\rangle \quad (3)$$

$$= e^{-iE_2 t} (-\sin \theta |\nu_e\rangle + \cos \theta |\nu_\mu\rangle), \quad (4)$$

where θ is the mixing angle. Show that the probability of finding a solar neutrino as an electron neutrino after a time T is given by

$$P = 1 - \sin^2(2\theta) \sin^2\left(\frac{(E_2 - E_1)T}{2}\right). \quad (5)$$

3. Take the high-energy limit $E \gg m_\nu$ to show that the probability of finding a solar neutrino with energy E as an electron neutrino at a distance L is given by

$$P = 1 - \sin^2(2\theta) \sin^2\left(\frac{\Delta m^2 L}{4E}\right). \quad (6)$$

4. How far should you put your detector from a reactor producing ~ 4 MeV neutrinos assuming $\Delta m^2 = 7.5 \times 10^{-5} \text{ eV}^2$ to see the largest effect?

Solution. Neglecting the thermal kinetic energy of the particles, the four-momentum before reaction is given by $k^\mu = (2m_p + m_e = s, 0)$. By four-momentum conservation $p_{\nu_e} + p_d = k$, we have

$$p_d^2 = m_d^2 = (k - p_{\nu_e})^2 = s + m_{\nu_e}^2 - 2\sqrt{s}E_{\nu_e} \implies E_{\nu_e} = \frac{s + m_{\nu_e}^2 - m_d^2}{2\sqrt{s}} = \sqrt{m_d^2 + |\mathbf{p}_{\nu_e}|^2}. \quad (7)$$

Therefore, we find

$$|\mathbf{p}_{\nu_e}| \approx 1.44 \text{ MeV}. \quad (8)$$

The solar neutrino is produced as an electron neutrino. That is, at $t = 0$,

$$|\nu; t = 0\rangle = |\nu_e\rangle = \cos \theta |\nu_1\rangle - \sin \theta |\nu_2\rangle. \quad (9)$$

Carrying out a standard QM calculation, it is easy to obtain (5).

In the high-energy limit,

$$E_1 \approx |\mathbf{p}| \left(1 - \frac{m_1^2}{2|\mathbf{p}|^2}\right) \implies -\Delta E \approx \frac{2\Delta m^2}{2|\mathbf{p}|} \approx \frac{2\Delta m^2}{2E}. \quad (10)$$

In natural units, $T \approx L$, and hence, we find the desired equation.

The maxima occur at

$$L_n = \frac{2\pi E}{\Delta m^2} (2n + 1), \quad L_0 \approx 66 \text{ km}. \quad (11)$$

6. Problem 29.8: Integrating out right-handed neutrinos

Show that when you integrate out the right-handed neutrinos in Eq. (29.63), a dimension-5 operator like that in Eq. (29.65) results. What is the exact relationship between M_{ij} and $\widetilde{M}_{ij}$?

Solution. The relevant Lagrangian for the right-handed neutrino is

$$\mathcal{L} = i\bar{\nu}_R^i(\not{\partial} - ig'Y_\nu\not{B})\nu_R^i - Y_{ij}^\nu\bar{L}^i\tilde{H}\nu_R^j - iM_{ij}(\nu_R^i)^c\nu_R^j, \quad (1)$$

where the hypercharge Y_ν vanishes. The equation of motion is given by

$$-i(\partial_\mu\nu_R^i)^\dagger\sigma^\mu - Y_{ji}^\nu(\bar{L}^j\tilde{H}) - 2iM_{ji}(\nu_R^j)^T\sigma_2 = 0. \quad (2)$$

At low energy compared to M_{ij} , the kinetic term of the right-handed neutrino can be neglected, and we find

$$-iM_{ij}(\nu_R^j)^T\sigma_2 = -\frac{1}{2}Y_{ji}^\nu(\bar{L}^j\tilde{H}), \quad M_{ij}\nu_R^j = -\frac{1}{2}i\sigma_2Y_{ij}^\nu(\bar{L}^j\tilde{H})^T, \quad (3)$$

where we have used $M_{ij} = M_{ji}$. Therefore, we find

$$-iM_{ij}(\nu_R^i)^c\nu_R^j = \frac{i}{4}Y_{ij}^\nu M_{jk}^{-1}Y_{kl}^\nu(\bar{L}^l\tilde{H})\sigma_2(\bar{L}^j\tilde{H})^T, \quad (4)$$

which gives

$$\boxed{\widetilde{M}_{il} = -\frac{i}{4}Y_{ij}^\nu M_{jk}^{-1}Y_{kl}^\nu.} \quad (5)$$

7. Problem 29.9: Chiral rotations and the θ angle

Show that when multiple generations are rotated, then the θ angle shifts by $\arg \det(R^\dagger L)$.

Solution. For N generations, the chiral rotations are

$$\psi_R^i \rightarrow R^{ij}\psi_R^j, \quad \psi_L^i \rightarrow L^{ij}\psi_L^j, \quad (1)$$

where $R, L \in U(N)$. We decompose

$$U(N) = SU(N) \times U(1), \quad (2)$$

and write

$$R = e^{ir/N}\tilde{R}, \quad L = e^{i\ell/N}\tilde{L}, \quad (3)$$

where

$$\tilde{R}, \tilde{L} \in SU(N), \quad \det \tilde{R} = \det \tilde{L} = 1. \quad (4)$$

Under the $SU(N)$ transformations, the Grassmann measures transform as

$$\prod_i \mathcal{D}\psi_R^i \rightarrow (\det \tilde{R})^{-1} \prod_i \mathcal{D}\psi_R^i = \prod_i \mathcal{D}\psi_R^i \quad (5)$$

and

$$\prod_i \mathcal{D}\psi_L^i \rightarrow (\det \tilde{L})^{-1} \prod_i \mathcal{D}\psi_L^i = \prod_i \mathcal{D}\psi_L^i. \quad (6)$$

The conjugate-field measures are also invariant. Therefore, the $SU(N)$ parts of R and L do not change the complete functional measure. Only the $U(1)$ parts can shift the θ angle.

The remaining $U(1)$ transformations are

$$\psi_R \rightarrow e^{ir/N}\psi_R, \quad \psi_L \rightarrow e^{i\ell/N}\psi_L. \quad (7)$$

These can be written as a vector rotation and an axial rotation:

$$\psi \rightarrow e^{i\alpha_V} e^{i\alpha_A\gamma^5} \psi, \quad (8)$$

where

$$\alpha_V = \frac{r + \ell}{2N}, \quad \alpha_A = \frac{r - \ell}{2N}. \quad (9)$$

The vector rotation leaves the functional measure invariant. An axial rotation by α_A shifts the θ angle by $-2\alpha_A$ for each generation. Since there are N generations,

$$\Delta\theta = -2N\alpha_A = -2N \left(\frac{r - \ell}{2N} \right) = \ell - r. \quad (10)$$

Finally,

$$\begin{aligned} \det(R^\dagger L) &= (\det R)^* \det L \\ &= e^{-ir} e^{i\ell} \\ &= e^{i(\ell - r)}. \end{aligned} \quad (11)$$

Therefore,

$$\arg \det(R^\dagger L) = \ell - r = \Delta\theta, \quad (12)$$

and hence

$$\boxed{\theta \rightarrow \theta + \arg \det(R^\dagger L)}. \quad (13)$$